

MASIRAH, Oman

WMO: 412880

Lat: 20.6714N

Lon: 58.8889E

Elev: 62

StdP: 14.66

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	64.9	66.7	42.9	40.8	75.9	46.7	47.3	75.4	24.6	68.8	22.6	69.8	10.1	300	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.9	98.5	75.8	95.7	76.4	93.6	77.2	82.8	89.9	81.8	88.9	81.0	88.2	12.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.0	161.3	86.7	80.1	156.7	86.2	79.0	151.1	85.7	47.0	89.9	45.8	89.3	45.0	88.5	94.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.3	24.8	22.8	DB	59.2	105.6	2.8	2.4	57.2	107.4	55.6	108.8	54.0	110.1	52.0	111.9	(4)
(5)			WB	52.0	84.5	2.6	2.9	50.1	86.7	48.6	88.4	47.1	90.0	45.2	92.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	80.9	73.3	74.4	78.8	84.8	88.3	86.8	82.4	80.8	81.7	82.9	80.0	75.9	(6)
(7)		DBStd	5.10	2.36	2.69	3.18	3.23	2.33	2.58	2.10	1.65	1.84	2.27	2.01	2.36	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11258	721	684	892	1043	1186	1104	1005	955	950	1018	899	801	(10)
(11)		CDD65	5783	256	264	427	593	721	654	540	490	500	553	449	336	(11)
(12)		CDH74	56389	883	1263	3515	6969	9569	8285	5313	4132	4718	5874	4036	1832	(12)
(13)		CDH80	20910	38	133	997	3082	5126	4073	1719	1080	1380	2125	984	173	(13)
(14)	Wind	WSAvg	11.6	8.9	8.7	8.7	10.5	13.9	15.6	18.0	16.1	12.0	8.1	8.5	9.5	(14)
(15)	Precipitation	PrecAvg	1.2	0.1	0.2	0.3	0.3	0.0	0.1	0.0	0.0	0.0	0.1	0.1	0.1	(15)
(16)		PrecMax	5.0	1.6	2.8	3.2	3.2	0.2	1.3	0.9	1.8	0.2	1.7	2.8	0.9	(16)
(17)		PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	1.5	0.3	0.5	0.6	0.7	0.0	0.3	0.1	0.2	0.0	0.3	0.4	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.5	86.5	93.7	100.6	103.0	101.5	94.7	90.9	92.4	95.6	89.8	84.6	(19)
(20)			MCWB	67.8	67.7	68.8	71.8	75.1	75.8	77.7	77.2	76.4	71.4	71.3	71.0	(20)
(21)		2%	DB	80.8	83.3	89.6	96.8	99.0	97.0	91.4	88.0	89.6	92.6	87.3	82.8	(21)
(22)			MCWB	68.9	69.1	69.9	71.7	76.2	78.7	78.0	76.9	76.6	72.6	73.6	70.9	(22)
(23)		5%	DB	79.3	81.3	87.1	93.6	96.0	94.2	89.1	86.4	87.6	90.1	86.0	81.9	(23)
(24)			MCWB	68.2	69.9	71.5	73.4	77.8	79.5	77.6	76.7	76.7	73.9	73.9	70.7	(24)
(25)		10%	DB	78.3	80.1	85.0	91.0	93.6	92.0	87.1	85.2	86.1	88.0	84.7	80.8	(25)
(26)			MCWB	67.6	69.7	71.9	74.6	78.8	80.0	77.3	76.3	76.4	75.1	73.4	70.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.2	75.9	78.2	83.1	84.7	84.6	80.9	79.7	79.2	80.8	78.8	76.5	(27)
(28)			MCDWB	78.7	80.5	84.5	89.3	92.7	91.9	88.3	86.3	87.2	86.0	85.2	81.1	(28)
(29)		2%	WB	72.3	73.8	76.6	80.6	82.7	83.1	79.5	78.3	78.2	79.2	77.4	74.2	(29)
(30)			MCDWB	77.6	79.5	83.2	87.1	90.2	90.1	87.4	85.3	86.2	85.0	83.5	80.0	(30)
(31)		5%	WB	70.9	72.6	75.7	79.4	81.7	82.0	78.6	77.4	77.5	78.4	76.4	72.8	(31)
(32)			MCDWB	76.8	78.5	82.2	86.1	89.3	88.9	86.4	84.5	85.1	84.4	82.8	79.2	(32)
(33)		10%	WB	69.5	71.6	74.7	78.4	81.0	81.3	77.8	76.7	76.9	77.7	75.3	71.8	(33)
(34)			MCDWB	76.1	77.5	81.2	85.3	88.6	88.2	85.3	83.6	84.1	83.9	82.2	78.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.8	11.9	13.1	14.0	13.9	12.8	11.5	11.1	11.2	12.6	10.4	10.0	(35)
(36)			MCDBR	11.6	13.4	15.6	17.3	16.9	16.0	14.2	12.3	12.7	15.2	11.6	10.8	(36)
(37)			MCWBR	6.2	7.2	8.3	8.6	6.5	5.0	4.5	4.5	4.1	6.9	5.7	5.3	(37)
(38)		5% WB	MCDWB	11.4	11.9	13.5	13.9	14.0	12.6	12.3	11.5	11.7	11.9	11.1	10.5	(38)
(39)			MCWBR	7.0	7.3	7.3	7.5	5.4	4.7	4.6	4.6	4.3	5.5	5.9	5.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.406	0.441	0.488	0.547	0.609	0.741	0.815	0.748	0.597	0.456	0.432	0.410	(40)	
(41)		taud	2.261	2.135	1.991	1.874	1.764	1.545	1.464	1.556	1.798	2.117	2.198	2.248	(41)	
(42)		Ebn at Noon	273	270	261	247	230	200	186	199	230	262	262	267	(42)	
(43)		Edh at Noon	40	48	57	66	73	89	97	89	69	48	42	39	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1567	1816	2092	2291	2368	2182	1981	1948	2017	1897	1615	1476	(44)	
(45)		RadStd	39	47	67	54	29	147	151	95	70	40	51	53	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-1.13	-1.78	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	-1.13	-1.78	N/A	N/A	N/A	N/A

Nomenclature: See separate page